



SANIDHYA KUMAR TIWARI

Roll No.: 2511AI31

M.TECH. (Artificial Intelligence)

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech. (AI)	Indian Institute of Technology, Patna	81.6%	2025–Present
B.Tech. (CSE)	Gautam Buddha University	87.7%	2019–2023

EXPERIENCE

- **Teaching Assistant, IIT Patna** July 2025 – May 2026
 - Assisted in tutorials and lab sessions on Data Structures, Algorithms, and C++ programming; mentored students in debugging, algorithm complexity analysis, and OOP concepts.
 - Evaluated programming assignments and provided actionable feedback on code quality, correctness, optimization.

PROJECTS

- **Network Deep Packet Inspection Engine** June 2026
High-performance network traffic analysis engine for inspecting and filtering packets from PCAP captures. GitHub
 - **Technologies:** C++, Linux, libpcap, TCP/IP, Ethernet, IPv4, TCP, UDP, TLS, Multithreading
 - Designed and developed a Deep Packet Inspection engine in C++ to analyze PCAP traffic and identify application-layer protocols.
 - Implemented packet parsing for Ethernet, IPv4, TCP, UDP, and TLS SNI extraction to classify encrypted HTTPS traffic.
 - Designed a multithreaded producer-consumer packet processing pipeline and implemented rule-based traffic filtering using IP, domain, and application policies.
- **Mini Unix Shell(PosixLite)** Feb 2026
Unix-like interactive shell built in C++ using Linux POSIX system calls. GitHub
 - **Technologies:** C++, Linux, POSIX System Calls, Process Management, Inter-Process Communication (IPC), File Descriptors
 - Designed and developed a Unix-like interactive shell in C++ using Linux POSIX APIs, demonstrating strong understanding of Operating Systems and System Programming.
 - Implemented process creation and execution using `fork()`, `execvp()`, `waitpid()`, pipes, and signals to support foreground and background process execution.
 - Developed multi-stage pipelines and file descriptor redirection using `dup2()`, showcasing Linux process communication and efficient file management.
 - Built a command parser supporting built-in commands (`cd`, `history`, `jobs`, `fg`, `exit`) using efficient data structures and object-oriented design principles.

TECHNICAL SKILLS

- **Languages:** C++, SQL
- **Core CS:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems, Computer Networks
- **Tools:** Git, Linux, VS Code

ACHIEVEMENTS

- **GATE Qualified, CSE** 2025
- **Secured 2nd Position**, B.Tech. CSE Department, Gautam Buddha University 2023
- **Gold Medal**, Shauryoustasva Sports Meet organized by GBU 2022
- **Silver Medal**, Regional Sport Meet organized by KVS 2015

CERTIFICATIONS

- NPTEL Certification — Database Management System
- Coding Ninjas Certification — Data Structures & Algorithms (C++)

POSITIONS OF RESPONSIBILITY

- **Placement Coordinator**, Training and Placement Cell, IIT Patna 2025 – 2027
- **Sports Coordinator**, Shauryoustasva, GBU Jan 2022 – Jan 2023
- **Scout Leader**, Scout and Guide, KVS Jun 2016 – Jul 2016